

packages/graph-view/src/utils/elk-bundled.ts

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T05:46:22.370Z

Overview

New file `packages/graph-view/src/utils/elk-bundled.ts` (lines 1-35) adds an ESM-compatible shim for the ELK layout engine's UMD bundle, enabling lazy loading in Vite/Electron contexts.

Key changes

- **Exported types**
 - `ElkLayoutInstance` (lines 3-5) – interface with `layout(graph: unknown): Promise<unknown>`.
 - `ElkConstructor` (line 7) – constructor type returning `ElkLayoutInstance`.
- `resolveElkConstructor` (lines 9-14) checks `mod.default` or `mod.ELK`; throws if neither is a function.
- `loadElkConstructor` (lines 19-25) performs a single dynamic import of `"elkjs/lib/elk.bundled.js"` and caches the resulting promise (`ctorPromise`).
- `loadElk` (lines 30-35) creates an `ElkLayoutInstance` by invoking the cached constructor and caches the instance promise (`instancePromise`).

Impact

- **Correctness:** `resolveElkConstructor` validates the constructor before use, preventing silent failures.
- **Performance:** Caching (`ctorPromise`, `instancePromise`) avoids repeated bundle loads, reducing startup time and memory usage.
- **Maintainability:** Centralizes ELK loading logic; other modules can `import { loadElk }` without handling UMD quirks.
- **Compatibility:** The file is a pure TS module with no side effects, suitable for both browser and Node environments.

Risks & follow-ups

- Runtime failure if `"elkjs/lib/elk.bundled.js"` is missing or corrupted; callers should handle promise rejection.
- Verify that the caching logic correctly prevents multiple imports across the application.
- Ensure that the returned `ElkLayoutInstance` satisfies downstream expectations.
- Test dynamic import behavior in Electron's preload script.